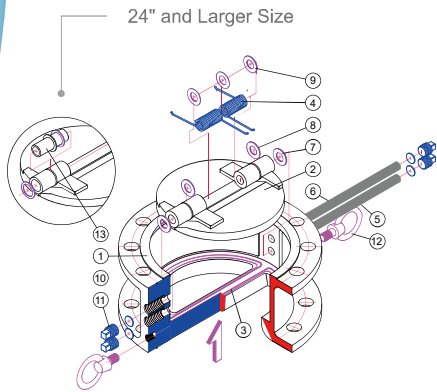


Flange Type

Valve Construction - Flange Type

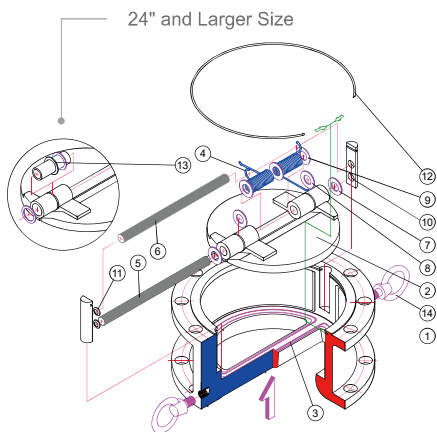
Products Range

Retainer Type (D Type)



Retainer Type (D Type)			
No.	Parts Name	Q'ty	Remarks
1	Body	1	
2	Plate	2	
3	Seat	1	
4	Spring	1	1-1/2"(40A) to 5"(125A)
		2	6"(150A) to 22"(550A)
		4	24"(600A) & Larger size
5	Hinge Pin	1	
6	Stop Pin	1	
7	Body Bearing	2	
8	Plate Bearing	2	
9	Spring Bearing	0	1-1/2"(40A) to 5"(125A)
		3	6"(150A) to 22"(550A)
		5	24"(600A) & Larger size
10	Stabilizer	4	For Rubber Seat Valve Only
11	Retainer	4	
12	Eye Bolt	1	8"(200A) & Larger size
13	Sleeve	2	24"(600A) & Larger size

Retainerless Type (I Type)



Retainerless Type (I Type)			
No.	Parts Name	Q'ty	Remarks
1	Body	1	
2	Plate	2	
3	Seat	1	
4	Spring	2	1-1/2"(40A) to 22"(550A)
		4	24"(600A) & Larger size
5	Hinge Pin	1	
6	Stop Pin	1	
7	Body Bearing	2	
8	Plate Bearing	2	
9	Spring Bearing	3	1-1/2"(40A) to 22"(550A)
		5	24"(600A) & Larger size
10	Adjust Plate	2	
11	Spring Washer	2	
12	Fix Ring	1	
13	Sleeve	2	24"(600A) & Larger size
14	Eye Bolt	1	8"(200A) & Larger size

Products Range

Body Material		CI			DI			CS / DI			CS / SS / AS													
Size		JIS		ANSI	JIS		ANSI	ISO/DIN		JIS			ANSI						ISO / DIN					
Inch	mm	5K	10K	125LB	5K	10K	150LB	PN10	PN16	5K	10K	20K	150LB	300LB	600LB	900LB	1500LB	2500LB	PN10	PN16	PN25	PN40		
1-1/2"	40A	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
2"	50A	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
2-1/2"	65A	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
3"	80A	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
4"	100A	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
5"	125A	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
6"	150A	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
8"	200A	-	●	●	-	●	●	-	-	-	●	●	●	●	●	●	●	●	-	-	-	-		
10"	250A	-	●	●	-	●	●	-	-	-	●	●	●	●	●	●	●	●	-	-	-	-		
12"	300A	-	●	●	-	●	●	-	-	-	●	●	●	●	●	●	●	●	-	-	-	-		
14"	350A	-	●	●	-	●	●	-	-	-	●	●	●	●	●	●	●	●	-	-	-	-		
16"	400A	-	●	●	-	●	●	-	-	-	●	●	●	●	●	●	●	●	-	-	-	-		
18"	450A	-	●	●	-	●	●	-	-	-	●	●	●	●	●	●	●	●	-	-	-	-		
20"	500A	-	●	●	-	●	●	-	-	-	●	●	●	●	●	●	●	-	-	-	-	-		
22"	550A	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
24"	600A	-	●	●	-	●	●	-	-	-	●	●	●	●	●	●	●	-	-	-	-	-		
26"	650A	-	●	●	-	●	●	-	-	-	●	-	●	-	-	-	-	-	-	-	-	-		
28"	700A	-	●	●	-	●	●	-	-	-	●	●	●	●	●	●	-	-	-	-	-	-		
30"	750A	-	●	●	-	●	●	-	-	-	●	●	●	●	●	●	-	-	-	-	-	-		
32"	800A	-	●	●	-	●	●	-	-	-	●	●	●	●	●	●	-	-	-	-	-	-		
36"	900A	-	●	●	-	●	●	-	-	-	●	●	●	●	●	●	-	-	-	-	-	-		
40"	1000A	-	●	●	-	●	●	-	-	-	●	-	●	●	●	●	-	-	-	-	-	-		
42"	1050A	-	-	-	-	-	-	-	-	-	-	-	●	●	●	●	-	-	-	-	-	-		
48"	1200A	-	●	●	-	●	●	-	-	-	●	-	●	●	●	●	-	-	-	-	-	-		
54"	1350A	-	●	●	-	●	●	-	-	-	●	-	●	●	●	-	-	-	-	-	-	-		
56"	1400A	-	-	●	-	-	●	-	-	-	-	-	●	-	-	-	-	-	-	-	-	-		
60"	1500A	-	●	●	-	●	●	-	-	-	●	-	●	●	●	-	-	-	-	-	-	-		

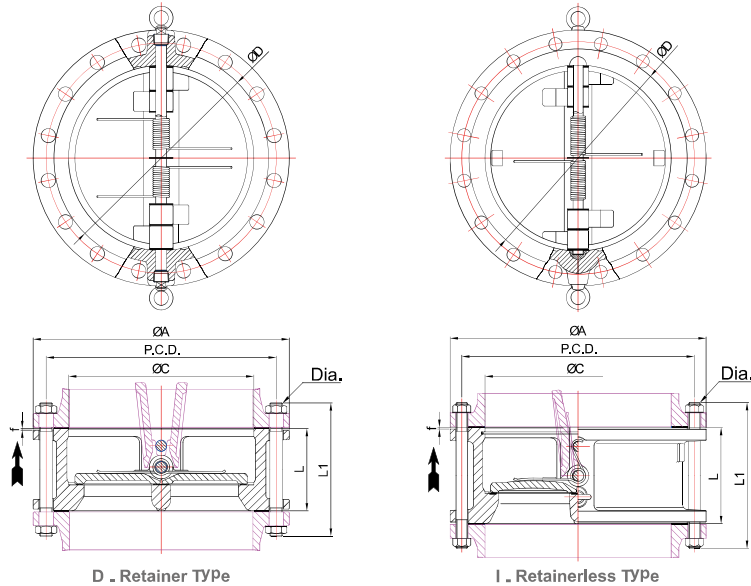
Note CI=Cast Iron ; DI=Ductile Iron ; CS=Carbon Steel ; SS=Stainless Steel ; AS=Alloy Steel

Flange Type

Products Range

Dual Plate Flange Type Check Valve

8" (200A) - 20" (500A)



Dual Plate Flange Type Check Valve

NPS	Class	Ø A	Ø C	L	P.C.D.	Ø D	f	Stud		L1		Approx. WT (kgs)
								No.	Dia.	FF/RF	RJ	
8" (200A)	* 10K	330	219	95	290.0	262	2	12	M20	230	-	38.0
	* 125/150LB	345	219	95	298.5	270	2	8	3/4"	250	-	42.0
	10K	330	219	127	290.0	262	2	12	M20	230	-	43.0
	20K	350	219	127	305.0	275	2	12	M22	250	-	65.0
	150LB	345	219	127	298.5	270	2	8	3/4"	250	-	46.0
	300LB	380	219	127	330.2	270	2	12	7/8"	280	-	70.0
	600LB	420	219	165	349.2	270	7	12	1-1/8"	370	385	105.0
	900LB	470	219	206	393.7	270	7	12	1-3/8"	440	455	180.0
	1500LB	485	219	206	393.7	270	7	12	1-5/8"	510	530	250.0
10" (250A)	* 10K	400	272	108	355.0	324	2	12	M22	255	-	75.0
	* 125/150LB	405	272	108	362.0	324	2	12	7/8"	280	-	82.0
	10K	400	272	146	355.0	324	2	12	M22	255	-	75.0
	20K	430	272	146	380.0	345	2	12	M24	280	-	98.0
	150LB	405	272	146	362.0	324	2	12	7/8"	280	-	82.0
	300LB	445	272	146	387.4	324	2	16	1"	320	-	105.0
	600LB	510	272	213	431.8	324	7	16	1-1/4"	440	455	165.0
	900LB	545	272	241	469.9	324	7	16	1-3/8"	485	505	220.0
	1500LB	585	272	248	482.6	324	7	12	1-7/8"	595	620	310.0
	2500LB	675	272	254	539.8	324	7	12	2-1/2"	750	785	420.0

Note * = Cast Iron & Ductile Iron

NPS	Class	Ø A	Ø C	L	P.C.D.	Ø D	f	Stud		L1		Approx. WT (kgs)
								No.	Dia.	FF/RF	RJ	
12" (300A)	* 10K	445	322	143	400.0	368	3	16	M22	270	-	82.0
	* 125/150LB	485	322	143	431.8	381	2	12	7/8"	280	-	88.0
	10K	445	322	181	400.0	368	3	16	M22	290	-	113.0
	20K	480	322	181	430.0	395	3	16	M24	315	-	140.0
	150LB	485	322	181	431.8	381	2	12	7/8"	315	-	125.0
	300LB	520	322	181	450.8	381	2	16	1-1/8"	375	-	153.0
	600LB	560	322	229	489.0	381	7	20	1-1/4"	465	480	239.0
	900LB	610	322	292	533.4	381	7	20	1-3/8"	555	575	347.0
	1500LB	675	322	305	571.5	381	7	16	2"	690	720	637.0
14" (350A)	2500LB	760	322	305	619.1	381	7	12	2-3/4"	850	885	747.0
	* 10K	490	356	184	445.0	413	3	16	M22	315	-	132.0
	* 125/150LB	535	356	184	476.3	413	2	12	1"	325	-	144.0
	10K	490	356	184	445.0	413	3	16	M22	300	-	132.0
	20K	540	356	222	480.0	440	3	16	M30	385	-	180.0
	150LB	535	356	184	476.3	413	2	12	1"	330	-	144.0
	300LB	585	356	222	514.4	413	2	20	1-1/8"	424	-	193.0
	600LB	605	356	273	527.0	413	7	20	1-3/8"	520	535	378.0
	900LB	640	356	356	558.8	413	7	20	1-1/2"	640	665	560.0
16" (400A)	1500LB	750	356	356	635.0	413	7	16	2-1/4"	775	805	1097.0
	* 10K	560	406	191	510.0	475	3	16	M24	325	-	162.0
	* 125/150LB	595	406	191	539.8	470	2	16	1"	340	-	176.0
	10K	560	406	191	510.0	475	3	16	M24	310	-	162.0
	20K	605	406	232	540.0	495	3	16	M30	410	-	280.0
	150LB	595	406	191	539.8	470	2	16	1"	340	-	176.0
	300LB	650	406	232	571.5	470	2	20	1-1/4"	445	-	301.0
	600LB	685	406	305	603.2	470	7	20	1-1/2"	570	585	451.0
	900LB	705	406	384	616.0	470	7	20	1-5/8"	680	705	547.0
18" (450A)	1500LB	825	406	384	704.8	470	7	16	2-1/2"	840	875	1255.0
	* 10K	620	457	203	565.0	530	3	20	M24	345	-	192.0
	* 125/150LB	635	457	203	577.9	533	2	16	1-1/8"	375	-	210.0
	10K	620	457	203	565.0	530	3	20	M24	325	-	192.0
	20K	675	457	264	605.0	560	3	20	M30	445	-	370.0
	150LB	635	457	203	577.9	533	2	16	1-1/8"	365	-	210.0
	300LB	710	457	264	628.6	533	2	24	1-1/4"	485	-	392.0
	600LB	745	457	362	654.0	533	7	20	1-5/8"	645	665	598.0
	900LB	785	457	451	685.8	533	7	20	1-7/8"	785	810	835.0
20" (500A)	1500LB	915	457	468	774.7	533	7	16	2-3/4"	970	1005	1775.0
	* 10K	675	508	213	620.0	585	3	20	M24	355	-	255.0
	* 125/150LB	700	508	213	635.0	584	2	20	1-1/8"	395	-	270.0
	10K	675	508	219	620.0	585	3	20	M24	345	-	249.0
	20K	730	508	292	660.0	615	3	20	M30	475	-	460.0
	150LB	700	508	219	635.0	584	2	20	1-1/8"	400	-	489.0
	300LB	775	508	292	685.8	584	2	24	1-1/4"	520	-	762.0
	600LB	815	508	368	723.9	584	7	24	1-5/8"	665	685	1783.0
	900LB	855	508	451	749.3	584	7	20	2"	805	830	2451.0

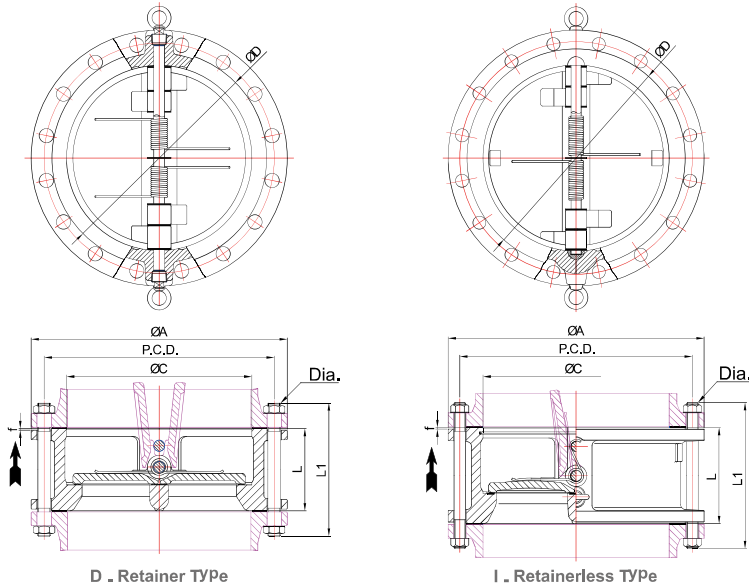
Note * = Cast Iron & Ductile Iron

Flange Type

Products Range

Dual Plate Flange Type Check Valve

24" (600A) - 60" (1500A)



Dual Plate Flange Type Check Valve

NPS	Class	Ø A	Ø C	L	P.C.D.	Ø D	f	Stud		L1		Approx. WT (kgs)
								No.	Dia.	FF/RF	RJ	
24" (600A)	* 10K	795	610	222	730.0	690	3	24	M30	395	-	354.0
	* 125/150LB	815	610	222	749.3	692	2	20	1-1/4"	415	-	381.0
	10K	795	610	222	730.0	690	3	24	M30	370	-	354.0
	20K	845	610	318	770.0	720	3	24	M36	520	-	361.0
	150LB	815	610	222	749.3	692	2	20	1-1/4"	420	-	381.0
	300LB	915	610	318	812.8	692	2	24	1-1/2"	570	-	880.0
	600LB	940	610	438	838.2	692	7	24	1-7/8"	775	795	1143.0
26" (650A)	900LB	1040	610	495	901.7	692	7	20	2-1/2"	940	970	1888.0
	* 10K	845	660	222	780.0	740	3	24	M30	455	-	943.0
	* 125/150LB	870	660	222	806.4	749	2	24	1-1/4"	470	-	1007.0
	10K	845	660	222	780.0	740	3	24	M30	510	-	943.0
28" (700A)	150LB	870	660	222	806.4	749	2	24	1-1/4"	595	-	1007.0
	* 10K	905	711	305	840.0	800	3	24	M30	500	-	965.0
	* 125/150LB	925	711	305	863.6	800	2	28	1-1/4"	515	-	1025.0
	10K	905	711	305	840.0	800	3	24	M30	465	-	965.0
	20K	995	711	368	900.0	840	5	24	M45	615	-	1100.0
	150LB	925	711	305	863.6	800	2	28	1-1/4"	565	-	1025.0
	300LB	1035	711	368	939.8	800	2	28	1-5/8"	670	-	1120.0
	600LB	1075	711	483	965.2	800	7	28	2"	845	870	1467.0

Note * = Cast Iron & Ductile Iron

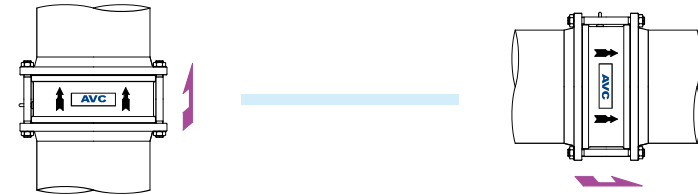
NPS	Class	Ø A	Ø C	L	P.C.D.	Ø D	f	Stud		L1		Approx. WT (kgs)
								No.	Dia.	FF/RF	RJ	
30" (750A)	* 10K	970	762	305	900.0	855	3	24	M30	490	-	744.0
	* 125/150LB	985	762	305	914.4	857	2	28	1-1/4"	515	-	793.0
	10K	970	762	305	900.0	855	3	24	M30	485	-	744.0
	20K	1080	762	368	970.0	900	5	24	M52	655	-	1570.0
	150LB	985	762	305	914.4	857	2	28	1-1/4"	595	-	793.0
	300LB	1090	762	368	997.0	857	2	28	1-3/4"	705	-	1603.0
	600LB	1130	762	505	1022.4	857	7	28	2"	870	900	2445.0
32" (800A)	* 10K	1020	813	356	950.0	905	3	28	M30	545	-	810.0
	* 125/150LB	1060	813	356	977.9	914	2	28	1-1/2"	570	-	844.0
	10K	1020	813	356	950.0	905	3	28	M30	510	-	810.0
	20K	1140	813	368	1030.0	960	5	24	M52	670	-	1860.0
	150LB	1060	813	356	977.9	914	2	28	1-1/2"	630	-	844.0
	300LB	1150	813	368	1054.1	914	2	28	1-7/8"	735	-	1935.0
	600LB	1195	813	533	1079.5	914	7	28	2-1/4"	920	945	2935.0
36" (900A)	* 10K	1120	914	368	1050.0	1005	3	28	M30	560	-	1080.0
	* 125/150LB	1170	914	368	1085.8	1022	2	32	1-1/2"	600	-	1148.0
	10K	1120	914	368	1050.0	1005	3	28	M30	555	-	1080.0
	20K	1250	914	483	1140.0	1070	5	28	M52	745	-	2090.0
	150LB	1170	914	368	1085.8	1022	2	32	1-1/2"	680	-	1148.0
	300LB	1270	914	483	1168.4	1022	2	32	2"	830	-	2140.0
	600LB	1315	914	635	1193.8	1022	7	28	2-1/2"	1045	-	4735.0
40" (1000A)	* 10K	1235	1016	432	1160.0	1110	3	28	M36	630	-	1470.0
	* 125/150LB	1290	1016	432	1200.2	1124	2	36	1-1/2"	645	-	1515.0
	10K	1235	1016	432	1160.0	1110	3	28	M36	595	-	1470.0
	150LB	1290	1016	432	1200.2	1124	2	36	1-1/2"	710	-	1515.0
	300LB	1240	1016	546	1155.7	1086	2	32	1-5/8"	895	-	3073.0
	600LB	1320	1016	660	1212.8	1111	7	32	2-1/4"	1130	-	6310.0
	* 125/150LB	1345	1067	432	1257.3	1194	2	36	1-1/2"	680	-	1650.0
42" (1050A)	150LB	1345	1067	432	1257.3	1194	2	36	1-1/2"	740	-	1650.0
	300LB	1290	1067	568	1206.5	1137	2	32	1-5/8"	925	-	4430.0
	600LB	1405	1067	702	1282.7	1168	7	28	2-1/2"	1200	-	5280.0
	* 10K	1465	1219	524	1380.0	1325	3	32	M36	750	-	2480.0
48" (1200A)	* 125/150LB	1510	1219	524	1422.4	1359	2	44	1-1/2"	775	-	2530.0
	10K	1465	1219	524	1380.0	1325	3	32	M36	705	-	3480.0
	150LB	1510	1219	524	1422.4	1359	2	44	1-1/2"	850	-	2530.0
	300LB	1465	1219	629	1371.6	1302	2	32	1-7/8"	1030	-	5010.0
54" (1350A)	600LB	1595	1219	787	1460.5	1334	7	32	2-3/4"	1240	-	6030.0
	* 10K	1630	1372	540	1540.0	1480	3	36	M42	780	-	-
	* 125/150LB	1685	1372	540	1593.8	1511	2	44	1-3/4"	820	-	-
	10K	1630	1372	540	1540.0	1480	3	36	M42	735	-	-
56" (1400A)	150LB	1685	1372	540	1593.8	1511	2	44	1-3/4"	905	-	-
	300LB	1660	1372	718	1549.4	1467	2	28	2-1/4"	1095	-	-
	* 125/150LB	1745	1422	568	1651.0	1575	2	48	1-3/4"	895	-	-
	150LB	1745	1422	568	1651.0	1575	2	48	1-3/4"	940	-	-
60" (1500A)	* 10K	1795	1524	660	1700.0	1635	3	40	M42	910	-	-
	* 125/150LB	1855	1524	660	1759.0	1676	2	52	1-3/4"	945	-	-
	10K	1795	1524	660	1700.0	1635	3	40	M42	860	-	-
	150LB	1855	1524	660	1759.0	1676	2	52	1-3/4"	1050	-	-
	300LB	1810	1524	838	1701.8	1626	2	32	2-1/4"	1315	-	-

Note * = Cast Iron & Ductile Iron

Best Practice Valve Installation

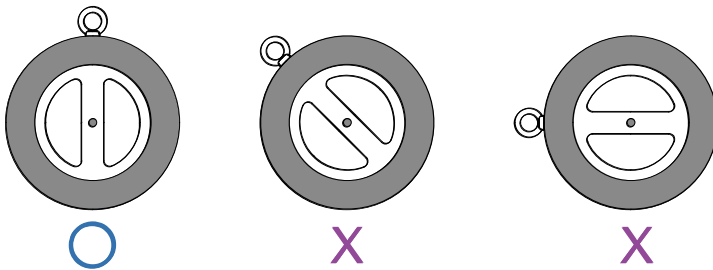
Installation Direction

Our wafer check valve can be installed in horizontal and vertical piping. Please install according to the casting arrow direction on body of our wafer check valve.



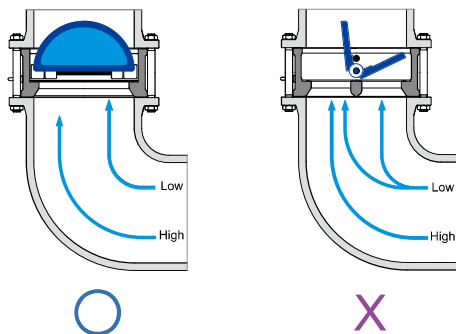
Horizontal Installation

For a horizontal installation make sure the check valve emboss must be in the vertical position if except eye-bolts or 5" & under.



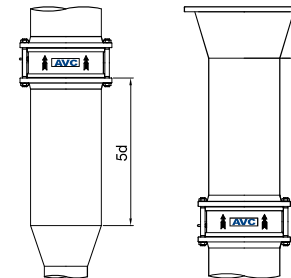
Vertical Installation

For elbow or pump exit installations, the valve rib is parallel to the incoming horizontal piping and creates an equal loading on the plates. Additional pressure drop should be expected due to the weight of the plates.

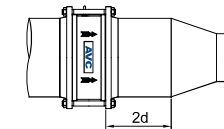


Distance of Installation

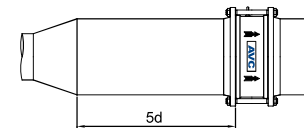
In order to avoid producing the turbulence, vibration and abrasion in the piping. We propose designing and installing according to the following distance.



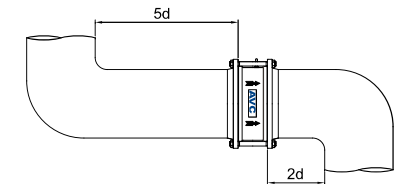
▲ 5 diameters minimum from reducer to valve and 5 diameters minimum after valve to expander.



▲ 2 diameters minimum after valve to reducer.



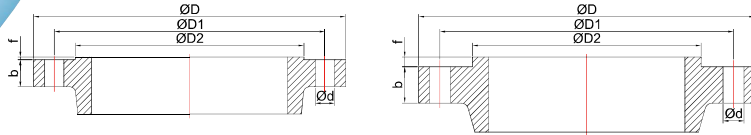
▲ 5 diameters minimum from reducer to valve.



▲ 5 diameters minimum from bend to valve and 2 diameters minimum after valve to bend.

Note "d" means diameter of the pipe.

Pipe Flange - ASME B16.5 RF



Class 150 to 300LB

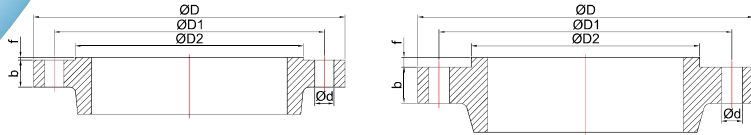
Class 600 to 1500LB

Class	NPS		Ø D	Ø D1	Ø D2	b	f	Ø d		Bolt	
	inch	mm						inch	mm	Number	Dia.
150LB	1/2	15	90	60.3	35	9.6	2	5/8	16	4	1/2
	3/4	20	100	69.9	43	11.2	2	5/8	16	4	1/2
	1	25	110	79.4	51	12.7	2	5/8	16	4	1/2
	1-1/4	32	115	88.9	64	14.3	2	5/8	16	4	1/2
	1-1/2	40	125	98.4	73	15.9	2	5/8	16	4	1/2
	2	50	150	120.7	92	17.5	2	3/4	19	4	5/8
	2-1/2	65	180	139.7	105	20.7	2	3/4	19	4	5/8
	3	80	190	152.4	127	22.3	2	3/4	19	4	5/8
	4	100	230	190.5	157	22.3	2	3/4	19	8	5/8
	5	125	255	215.9	186	22.3	2	7/8	22	8	3/4
	6	150	280	241.3	216	23.9	2	7/8	22	8	3/4
	8	200	345	298.5	270	27.0	2	7/8	22	8	3/4
	10	250	405	362.0	324	28.6	2	1	25	12	7/8
	12	300	485	431.8	381	30.2	2	1	25	12	7/8
	14	350	535	476.3	413	33.4	2	1-1/8	28	12	1
	16	400	595	539.8	470	35.0	2	1-1/8	28	16	1
	18	450	635	577.9	533	38.1	2	1-1/4	32	16	1-1/8
	20	500	700	635.0	584	41.3	2	1-1/4	32	20	1-1/8
	24	600	815	749.3	692	46.1	2	1-3/8	35	20	1-1/4
300LB	1/2	15	95	66.7	35	12.7	2	5/8	16	4	1/2
	3/4	20	115	82.6	43	14.3	2	3/4	19	4	5/8
	1	25	125	88.9	51	15.9	2	3/4	19	4	5/8
	1-1/4	32	135	98.4	64	17.5	2	3/4	19	4	5/8
	1-1/2	40	155	114.3	73	19.1	2	7/8	22	4	3/4
	2	50	165	127.0	92	20.7	2	3/4	19	8	5/8
	2-1/2	65	190	149.2	105	23.9	2	7/8	22	8	3/4
	3	80	210	168.3	127	27.0	2	7/8	22	8	3/4
	4	100	255	200.0	157	30.2	2	7/8	22	8	3/4
	5	125	280	235.0	186	33.4	2	7/8	22	8	3/4
	6	150	320	269.9	216	35.0	2	7/8	22	12	3/4
	8	200	380	330.2	270	39.7	2	1	25	12	7/8
	10	250	445	387.4	324	46.1	2	1-1/8	28	16	1
	12	300	520	450.8	381	49.3	2	1-1/4	32	16	1-1/8
	14	350	585	514.4	413	52.4	2	1-1/4	32	20	1-1/8
	16	400	650	571.5	470	55.6	2	1-3/8	35	20	1-1/4
	18	450	710	628.6	533	58.8	2	1-3/8	35	24	1-1/4
	20	500	775	685.8	584	62.0	2	1-3/8	35	24	1-1/4
	24	600	915	812.8	692	68.3	2	1-5/8	41	24	1-1/2
400LB	1/2	15	95	66.7	35	12.7	2	5/8	16	4	1/2
	3/4	20	115	82.6	43	14.3	2	3/4	19	4	5/8
	1	25	125	88.9	51	15.9	2	3/4	19	4	5/8
	1-1/4	32	135	98.4	64	17.5	2	3/4	19	4	5/8
	1-1/2	40	155	114.3	73	19.1	2	7/8	22	4	3/4
	2	50	165	127.0	92	20.7	2	3/4	19	8	5/8
	2-1/2	65	190	149.2	105	23.9	2	7/8	22	8	3/4
	3	80	210	168.3	127	27.0	2	7/8	22	8	3/4
	4	100	255	200.0	157	30.2	2	7/8	22	8	3/4
	5	125	280	235.0	186	33.4	2	7/8	22	8	3/4
	6	150	320	269.9	216	35.0	2	7/8	22	12	3/4
	8	200	380	330.2	270	39.7	2	1	25	12	7/8
	10	250	445	387.4	324	46.1	2	1-1/8	28	16	1
	12	300	520	450.8	381	49.3	2	1-1/4	32	16	1-1/8
	14	350	585	514.4	413	52.4	2	1-1/4	32	20	1-1/8
	16	400	650	571.5	470	55.6	2	1-3/8	35	20	1-1/4
	18	450	710	628.6	533	58.8	2	1-3/8	35	24	1-1/4
	20	500	775	685.8	584	62.0	2	1-3/8	35	24	1-1/4
	24	600	915	812.8	692	68.3	2	1-5/8	41	24	1-1/2
	1/2	15	95	66.7	35	12.7	2	5/8	16	4	1/2
	3/4	20	115	82.6	43	14.3	2	3/4	19	4	5/8
	1	25	125	88.9	51	15.9	2	3/4	19	4	5/8
	1-1/4	32	135	98.4	64	17.5	2	3/4	19	4	5/8
	1-1/2	40	155	114.3	73	19.1	2	7/8	22	4	3/4
	2	50	165	127.0	92	20.7	2	3/4	19	8	5/8
	2-1/2	65	190	149.2	105	23.9	2	7/8	22	8	3/4
	3	80	210	168.3	127	27.0	2	7/8	22	8	3/4
	4	100	255	200.0	157	30.2	2	7/8	22	8	3/4
	5	125	280	235.0	186	33.4	2	7/8	22	8	3/4
	6	150	320	269.9	216	35.0	2	7/8	22	12	3/4
	8	200	380	330.2	270	39.7	2	1	25	12	7/8
	10	250	445	387.4	324	46.1	2	1-1/8	28	16	1
	12	300	520	450.8	381	49.3	2	1-1/4	32	16	1-1/8
	14	350	585	514.4	413	52.4	2	1-1/4	32	20	1-1/8
	16	400	650	571.5	470	55.6	2	1-3/8	35	20	1-1/4
	18	450	710	628.6	533	58.8	2	1-3/8	35	24	1-1/4
	20	500	775	685.8	584	62.0	2	1-3/8	35	24	1-1/4
	24	600	915	812.8	692	68.3	2	1-5/8	41	24	1-1/2

Use Class 600 dimensions in these sizes

Class	NPS		Ø D	Ø D1	Ø D2	b	f	Ø d		Bolt	
	inch	mm						inch	mm	Number	Dia.
600LB	1/2	15	95	66.7	35	14.3	7	5/8	16	4	1/2
	3/4	20	115	82.6	43	15.9	7	3/4	19	4	5/8
	1	25	125	88.9	51	17.5	7	3/4	19	4	5/8
	1-1/4	32	135	98.4	64	20.7	7	3/4	19	4	5/8
	1-1/2	40	155	114.3	73	22.3	7	7/8	22	4	3/4
	2	50	165	127.0	92	25.4	7	3/4	19	8	5/8
	2-1/2	65	190	149.2	105	28.6	7	7/8	22	8	3/4
	3	80	210	168.3	127	31.8	7	7/8	22	8	3/4
	4	100	275	215.9	157	38.1	7	1	25	8	7/8
	5	125	330	266.7	186	44.5	7	1-1/8	28	8	1
	6	150	355	292.1	216	47.7	7	1-1/8	28	12	1
	8	200	420	349.2	270	55.6	7	1-1/4	32	12	1-1/8
	10	250	510	431.8	324	63.5	7	1-3/8	35	16	1-1/4
	12	300	560	489.0	381	66.7	7	1-3/8	35	20	1-1/4
	14	350	605	527.0	413	69.9	7	1-1/2	38	20	1-3/8
	16	400	685	603.2	470	76.2	7	1-5/8	41	20	1-1/2
	18	450	745	654.0	533	82.6	7	1-3/4	44	20	1-5/8
	20	500	815	723.9	584	88.9	7	1-3/4	44	24	1-5/8
	24	600	940	838.2	692	101.6	7	2	51	24	1-7/8
	1/2	15	120	82.6	35	22.3	7	7/8	22	4	3/4
	3/4	20	130	88.9	43	25.4	7	7/8	22	4	3/4
	1	25	150	101.6	51	28.6	7	1	25	4	7/8
	1-1/4	32	160	111.1	64	28.6	7	1	25	4	7/8
	1-1/2	40	180	123.8	73	31.8	7	1-1/8	28	4	1
2	50	215	165.1	92	38.1	7	1	25	8	7/8	
2-1/2	65	245	190.5	105	41.3	7	1-1/8	28	8	1	
3	80	240	190.5	127	38.1	7	1	25	8	7/8	
4	100	290	235.0	157	44.5	7	1-1/4	32	8	1-1/8	
5	125	350	279.4	186	50.8	7	1-3/8	35	8	1-1/4	
6	150	380	317.5	216	55.6	7	1-1/4	32	12	1-1/8	
8	200	470	393.7	270	63.5	7	1-1/2	38	12	1-3/8	
10	250	545	469.9	324	69.9	7	1-1/2	38	16	1-3/8	
12	300	610	533.4	381	79.4	7	1-1/2	38	20	1-3/8	
14	350	640	558.8	413	85.8	7	1-5/8	41	20	1-1/2	
16	400	705	616.0	470	88.9	7	1-3/4	44	20	1-5/8	
18	450	785	685.8	533	101.6	7	2	51	20	1-7/8	
20	500	855	749.3	584	108.0	7	2-1/8	54	20	2	
24	600	1040	901.7	692	139.7	7	2-5/8	67	20	2-1/2	
1500LB	1/2	15	120	82.6	35	22.3	7	7/8	22	4	3/4
	3/4	20	130	88.9	43	25.4	7	7/8	22	4	3/4
	1	25	150	101.6	51	28.6	7	1	25	4	7/8
	1-1/4	32	160	111.1	64	28.6	7	1	25	4	7/8
	1-1/2	40	180	123.8	73	31.8	7	1-1/8	28	4	1
	2	50	215	165.1	92	38.1	7	1	25	8	7/8
	2-1/2	65	245	190.5	105	41.3	7	1-1/8	28	8	1
	3	80	265	203.2	127	47.7	7	1-1/4	32	8	1-1/8
	4	100	310	241.3	157	54.0	7	1-3/8	35	8	1-1/4
	5	125	375	292.1	186	73.1	7	1-5/8	41	8	1-1/2
	6	150	395	317.5	216	82.6	7	1-1/2	38	12	1-3/8
	8	200	485	393.7	270	92.1	7	1-3/4	44	12	1-5/8
	10	250	585	482.6	324	108.0	7	2	51	12	1-7/8
	12	300	675	571.5	381	123.9	7	2-1/8	54	16	2
	14	350	750	635.0	413	133.4	7	2-3/8	60	16	2-1/4
	16	400	825	704.8	470	146.1	7	2-5/8	67	16	2-1/2
	18	450	915	774.7	533	162.0	7	2-7/8	73	16	2-3/4
	20	500	985	831.8	584	177.8	7	3-1/8	79	16	3
	24	600	1170	990.6	692	203.2	7	3-5/8	92	16	3-1/2
	1/2	15	135	88.9	35	30.2	7	7/8	22	4	3/4
	3/4	20	140	95.2	43	31.8	7	7/8	22	4	3/4
	1	25	160	108.0	51	35.0	7	1	25	4	7/8
	1-1/4	32	185	130.2	64	38.1	7	1-1/8	28	4	1
	1-1/2	40	205	146.0	73	44.5	7	1-1/4	32	4	1-1/8
2	50	235	171.4	92	50.9	7	1-1/8	28	8	1	
2-1/2	65	265	196.8	105	57.2	7	1-1/4	32	8	1-1/8	
3	80	305	228.6	127	66.7	7	1-3/8	35	8	1-1/4	
4	100	355	273.0	157	76.2	7	1-5/8	41	8	1-1/2	
5	125	420	323.8	186	92.1	7	1-7/8	48	8	1-3/4	
6	150	485	368.3	216	108.0	7	2-1/8	54	8	2	
8	200	550	438.2	270	127.0	7	2-1/8	54	12	2	
10	250	675	539.8	324	165.1	7	2-5/8	67	12	2-1/2	
12	300	760	619.1	381	184.2	7	2-7/8	73	12	2-3/4	

Pipe Flange - ASME B16.47 Series A / MSS SP-44 RF

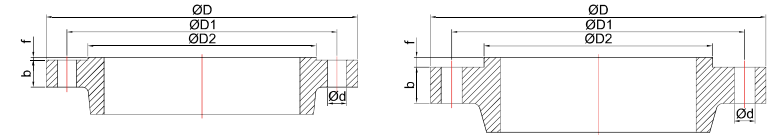


Class 150 to 300LB

Class 600 to 1500LB

Class	NPS		Ø D	Ø D1	Ø D2	b	f	Ø d		Bolt	
	inch	mm						inch	mm	Number	Dia.
150LB	26	650	870	806.4	749	66.7	2	1-3/8	35	24	1-1/4
	28	700	925	863.6	800	69.9	2	1-3/8	35	28	1-1/4
	30	750	985	914.4	857	73.1	2	1-3/8	35	28	1-1/4
	32	800	1060	977.9	914	79.4	2	1-5/8	41	28	1-1/2
	36	900	1170	1085.8	1022	88.9	2	1-5/8	41	32	1-1/2
	40	1000	1290	1200.2	1124	88.9	2	1-5/8	41	36	1-1/2
	42	1050	1345	1257.3	1194	95.3	2	1-5/8	41	36	1-1/2
	48	1200	1510	1422.4	1359	106.4	2	1-5/8	41	44	1-1/2
	54	1350	1685	1593.8	1511	119.1	2	1-7/8	48	44	1-3/4
	56	1400	1745	1651.0	1575	122.3	2	1-7/8	48	48	1-3/4
300LB	60	1500	1855	1759.0	1676	130.2	2	1-7/8	48	52	1-3/4
	26	650	970	876.3	749	77.8	2	1-3/4	44	28	1-5/8
	28	700	1035	939.8	800	84.2	2	1-3/4	44	28	1-5/8
	30	750	1090	997.0	857	90.5	2	1-7/8	48	28	1-3/4
	32	800	1150	1054.1	914	96.9	2	2	51	28	1-7/8
	36	900	1270	1168.4	1022	103.2	2	2-1/8	54	32	2
	40	1000	1240	1155.7	1086	112.8	2	1-3/4	44	32	1-5/8
	42	1050	1290	1206.5	1137	117.5	2	1-3/4	44	32	1-5/8
	48	1200	1465	1371.6	1302	131.8	2	2	51	32	1-7/8
	54	1350	1660	1549.4	1467	150.9	2	2-3/8	60	28	2-1/4
600LB	56	1400	1710	1600.2	1518	152.4	2	2-3/8	60	28	2-1/4
	60	1500	1810	1701.8	1626	162.0	2	2-3/8	60	32	2-1/4
	26	650	1015	914.4	749	108.0	7	2	51	28	1-7/8
	28	700	1075	965.2	800	111.2	7	2-1/8	54	28	2
	30	750	1130	1022.4	857	114.3	7	2-1/8	54	28	2
	32	800	1195	1079.5	914	117.5	7	2-3/8	60	28	2-1/4
	36	900	1315	1193.8	1022	123.9	7	2-5/8	67	28	2-1/2
	40	1000	1320	1212.8	1111	158.8	7	2-3/8	60	32	2-1/4
	42	1050	1405	1282.7	1168	168.3	7	2-5/8	67	28	2-1/2
	48	1200	1595	1460.5	1334	189.0	7	2-7/8	73	32	2-3/4
900LB	26	650	1085	952.5	749	139.7	7	2-7/8	73	20	2-3/4
	28	700	1170	1022.4	800	142.9	7	3-1/8	79	20	3
	30	750	1230	1085.8	857	149.3	7	3-1/8	79	20	3
	32	800	1315	1155.7	914	158.8	7	3-3/8	86	20	3-1/4
	36	900	1460	1289.0	1022	171.5	7	3-5/8	92	20	3-1/2

Pipe Flange - ASME B16.47 Series B / API 605 RF

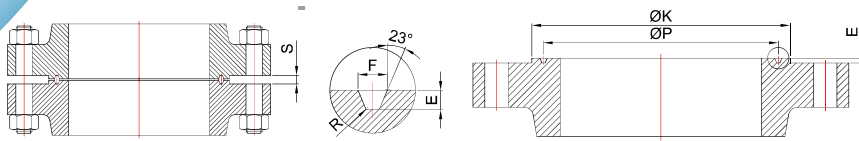


Class 150 to 300LB

Class 600 to 1500LB

Class	NPS		Ø D	Ø D1	Ø D2	b	f	Ø d		Bolt	
	inch	mm						inch	mm	Number	Dia.
150LB	26	650	785	744.5	711	39.8	2	7/8	22	36	3/4
	28	700	835	795.3	762	43.0	2	7/8	22	40	3/4
	30	750	885	846.1	813	43.0	2	7/8	22	44	3/4
	32	800	940	900.1	864	44.6	2	7/8	22	48	3/4
	36	900	1055	1009.6	972	50.9	2	1	25	44	7/8
	40	1000	1175	1120.8	1080	54.1	2	1-1/8	28	44	1
	42	1050	1225	1171.6	1130	57.3	2	1-1/8	28	48	1
	48	1200	1390	1335.1	1289	63.6	2	1-1/4	32	44	1-1/8
	54	1350	1550	1492.2	1441	70.0	2	1-1/4	32	56	1-1/8
	56	1400	1600	1543.0	1492	71.6	2	1-1/4	32	60	1-1/8
300LB	60	1500	1725	1662.1	1600	74.7	2	1-3/8	35	52	1-1/4
	26	650	865	803.3	737	87.4	2	1-3/8	35	32	1-1/4
	28	700	920	857.2	787	87.4	2	1-3/8	35	36	1-1/4
	30	750	990	920.8	845	92.1	2	1-1/2	38	36	1-3/8
	32	800	1055	977.9	902	101.6	2	1-5/8	41	32	1-1/2
	36	900	1170	1089.0	1010	101.6	2	1-3/4	44	32	1-5/8
	40	1000	1275	1190.6	1114	114.3	2	1-3/4	44	40	1-5/8
	42	1050	1335	1244.6	1168	117.5	2	1-7/8	48	36	1-3/4
	48	1200	1510	1416.0	1327	127.0	2	2	51	40	1-7/8
	54	1350	1675	1578.0	1480	135.0	2	2	51	48	1-7/8
600LB	56	1400	1765	1651.0	1537	152.4	2	2-3/8	60	36	2-1/4
	60	1500	1880	1763.7	1651	149.3	2	2-3/8	60	40	2-1/4
	26	650	890	806.4	727	111.2	7	1-3/4	44	28	1-5/8
	28	700	950	863.6	784	115.9	7	1-7/8	48	28	1-3/4
	30	750	1020	927.1	841	125.5	7	2	51	28	1-7/8
	32	800	1085	984.2	895	130.2	7	2-1/8	54	28	2
	36	900	1215	1104.9	1010	146.1	7	2-3/8	60	28	2-1/4
	26	650	1020	901.7	762	135.0	7	2-5/8	67	20	2-1/2
	28	700	1105	971.6	819	147.7	7	2-7/8	73	20	2-3/4
	30	750	1180	1035.0	876	155.6	7	3-1/8	79	20	3
900LB	32	800	1240	1092.2	927	160.4	7	3-1/8	79	20	3
	36	900	1345	1200.2	1029	173.1	7	3-1/8	79	24	3

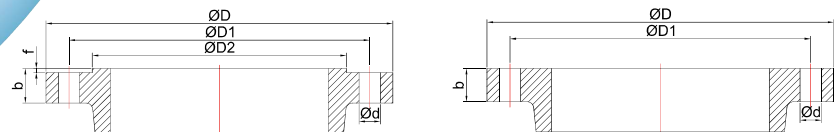
Pipe Flange - ASME B16.5 RTJ



Class	NPS		Ring Number	Ø P	E	F	R	Ø K	S
	inch	mm							
150LB	1	25	R15	47.63	6.35	8.74	0.8	63.5	4
	1-1/4	32	R17	57.15	6.35	8.74	0.8	73.0	4
	1-1/2	40	R19	65.07	6.35	8.74	0.8	82.5	4
	2	50	R22	82.55	6.35	8.74	0.8	102.0	4
	2-1/2	65	R25	101.60	6.35	8.74	0.8	121.0	4
	3	80	R29	114.30	6.35	8.74	0.8	133.0	4
	4	100	R36	149.23	6.35	8.74	0.8	171.0	4
	5	125	R40	171.45	6.35	8.74	0.8	194.0	4
	6	150	R43	193.68	6.35	8.74	0.8	219.0	4
	8	200	R48	247.65	6.35	8.74	0.8	273.0	4
	10	250	R52	304.80	6.35	8.74	0.8	330.0	4
	12	300	R56	381.00	6.35	8.74	0.8	406.0	4
	14	350	R59	396.88	6.35	8.74	0.8	425.0	3
	16	400	R64	454.03	6.35	8.74	0.8	483.0	3
300/600LB	18	450	R68	517.53	6.35	8.74	0.8	546.0	3
	20	500	R72	558.80	6.35	8.74	0.8	597.0	3
	24	600	R76	673.10	6.35	8.74	0.8	711.0	3
	1/2	15	R11	34.14	5.54	7.14	0.8	51.0	3
	3/4	20	R13	42.88	6.35	8.74	0.8	63.5	4
	1	25	R16	50.80	6.35	8.74	0.8	70.0	4
	1-1/4	32	R18	60.33	6.35	8.74	0.8	79.5	4
	1-1/2	40	R20	68.27	6.35	8.74	0.8	90.5	4
	2	50	R23	82.55	7.92	11.91	0.8	108.0	6 5
	2-1/2	65	R26	101.60	7.92	11.91	0.8	127.0	6 5
	3	80	R31	123.83	7.92	11.91	0.8	146.0	6 5
	4	100	R37	149.23	7.92	11.91	0.8	175.0	6 5
	5	125	R41	180.98	7.92	11.91	0.8	210.0	6 5
	6	150	R45	211.12	7.92	11.91	0.8	241.0	6 5
	8	200	R49	269.88	7.92	11.91	0.8	302.0	6 5
	10	250	R53	323.85	7.92	11.91	0.8	356.0	6 5
	12	300	R57	381.00	7.92	11.91	0.8	413.0	6 5
	14	350	R61	419.10	7.92	11.91	0.8	457.0	6 5
	16	400	R65	469.90	7.92	11.91	0.8	508.0	6 5
	18	450	R69	533.40	7.92	11.91	0.8	575.0	6 5
	20	500	R73	584.20	9.53	13.49	1.5	635.0	6 5
	24	600	R77	692.15	11.13	16.66	1.5	749.0	6

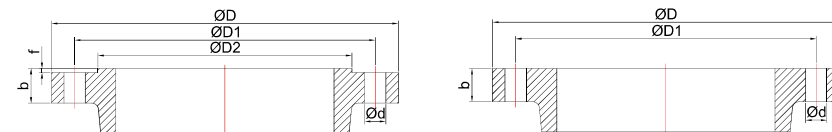
Class	NPS		Ring Number	Ø P	E	F	R	Ø K	S
	inch	mm							
900LB	1/2	15	R12	39.67	6.35	8.74	0.8	60.5	4
	3/4	20	R14	44.45	6.35	8.74	0.8	66.5	4
	1	25	R16	50.80	6.35	8.74	0.8	71.5	4
	1-1/4	32	R18	60.33	6.35	8.74	0.8	81.0	4
	1-1/2	40	R20	68.27	6.35	8.74	0.8	92.0	4
	2	50	R24	95.25	7.92	11.91	0.8	124.0	3
	2-1/2	65	R27	107.95	7.92	11.91	0.8	137.0	3
	3	80	R31	123.83	7.92	11.91	0.8	156.0	4
	4	100	R37	149.23	7.92	11.91	0.8	181.0	4
	5	125	R41	180.98	7.92	11.91	0.8	216.0	4
	6	150	R45	211.12	7.92	11.91	0.8	241.0	4
	8	200	R49	269.88	7.92	11.91	0.8	308.0	4
	10	250	R53	323.85	7.92	11.91	0.8	362.0	4
	12	300	R57	381.00	7.92	11.91	0.8	419.0	4
	14	350	R62	419.10	11.13	16.66	1.5	467.0	4
	16	400	R66	469.90	11.13	16.66	1.5	524.0	4
	18	450	R70	533.40	12.70	19.84	1.5	594.0	5
	20	500	R74	584.20	12.70	19.84	1.5	648.0	5
	24	600	R78	692.15	15.88	26.97	2.4	772.0	6
	1/2	15	R12	39.67	6.35	8.74	0.8	60.5	4
	3/4	20	R14	44.45	6.35	8.74	0.8	66.5	4
	1	25	R16	50.80	6.35	8.74	0.8	71.5	4
	1-1/4	32	R18	60.33	6.35	8.74	0.8	81.0	4
	1-1/2	40	R20	68.27	6.35	8.74	0.8	92.0	4
	2	50	R24	95.25	7.92	11.91	0.8	124.0	3
1500LB	2-1/2	65	R27	107.95	7.92	11.91	0.8	137.0	3
	3	80	R35	136.53	7.92	11.91	0.8	168.0	3
	4	100	R39	161.93	7.92	11.91	0.8	194.0	3
	5	125	R44	193.68	7.92	11.91	0.8	229.0	3
	6	150	R46	211.14	9.53	13.49	1.5	248.0	3
	8	200	R50	269.88	11.13	16.66	1.5	318.0	4
	10	250	R54	323.85	11.13	16.66	1.5	371.0	4
	12	300	R58	381.00	14.27	23.01	1.5	438.0	5
	14	350	R63	419.10	15.88	26.97	2.4	489.0	6
	16	400	R67	469.90	17.48	30.18	2.4	546.0	8
	18	450	R71	533.40	17.48	30.18	2.4	613.0	8
	20	500	R75	584.20	17.48	33.32	2.4	673.0	10
	24	600	R79	692.15	20.62	36.53	2.4	794.0	11
	1/2	15	R13	42.88	6.35	8.74	0.8	65.0	4
2500LB	3/4	20	R16	50.80	6.35	8.74	0.8	73.0	4
	1	25	R18	60.33	6.35	8.74	0.8	82.5	4
	1-1/4	32	R21	72.23	7.92	11.91	0.8	102.0	3
	1-1/2	40	R23	82.55	7.92	11.91	0.8	114.0	3
	2	50	R26	101.60	7.92	11.91	0.8	133.0	3
	2-1/2	65	R28	111.13	9.53	13.49	1.5	149.0	3
	3	80	R32	127.00	9.53	13.49	1.5	168.0	3
	4	100	R38	157.18	11.13	16.66	1.5	203.0	4
	5	125	R42	190.50	12.70	19.84	1.5	241.0	4
	6	150	R47	228.60	12.70	19.84	1.5	279.0	4
	8	200	R51	279.40	14.27	23.01	1.5	340.0	5
	10	250	R55	342.90	17.48	30.18	2.4	425.0	6
	12	300	R60	406.40	17.48	33.32	2.4	495.0	8

Pipe Flange - JIS B2220 RF / RF



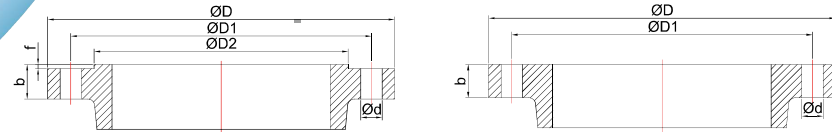
Class	NPS		Ø D	Ø D1	Ø D2	b	f	Bolt	
	inch	mm						Ø d	Number
5K	1/2	15	80	60	44	9	1	12	4
	3/4	20	85	65	49	10	1	12	4
	1	25	95	75	59	10	1	12	4
	1-1/4	32	115	90	70	12	2	15	4
	1-1/2	40	120	95	75	12	2	15	4
	2	50	130	105	85	14	2	15	4
	2-1/2	65	155	130	110	14	2	15	4
	3	80	180	145	121	14	2	19	4
	4	100	200	165	141	16	2	19	8
	5	125	235	200	176	16	2	19	8
	6	150	265	230	206	18	2	19	8
	8	200	320	280	252	20	2	23	8
	10	250	385	345	317	22	2	23	12
	12	300	430	390	360	22	3	23	12
	14	350	480	435	403	24	3	25	12
	16	400	540	495	463	24	3	25	16
	18	450	605	555	523	24	3	25	16
	20	500	655	605	573	24	3	25	20
	22	550	720	665	630	26	3	27	20
	24	600	770	715	680	26	3	27	20
	26	650	825	770	735	26	3	27	24
	28	700	875	820	785	26	3	27	24
	30	750	945	880	840	28	3	33	24
	32	800	995	930	890	28	3	33	24
	36	900	1095	1030	990	30	3	33	24
	40	1000	1195	1130	1090	32	3	33	28
	44	1100	1305	1240	1200	32	3	33	28
	48	1200	1420	1350	1305	34	3	33	32
	54	1350	1575	1505	1460	34	3	33	32
	60	1400	1730	1660	1615	36	3	33	36
10K	1/2	15	95	70	51	12	1	15	4
	3/4	20	100	75	56	14	1	15	4
	1	25	125	90	67	14	1	19	4
	1-1/4	32	135	100	76	16	2	19	4
	1-1/2	40	140	105	81	16	2	19	4
	2	50	155	120	96	16	2	19	4
	2-1/2	65	175	140	116	18	2	19	4
	3	80	185	150	126	18	2	19	8
	4	100	210	175	151	18	2	19	8
	5	125	250	210	182	20	2	23	8
	6	150	280	240	212	22	2	23	8
	8	200	330	290	262	22	2	23	12
	10	250	400	355	324	24	2	25	12
	12	300	445	400	368	24	3	25	16
	14	350	490	445	413	26	3	25	16
	16	400	560	510	475	28	3	27	16
	18	450	620	565	530	30	3	27	20
	20	500	675	620	585	30	3	27	20
	22	550	745	680	640	32	3	33	20
	24	600	795	730	690	32	3	33	24
	26	650	845	780	740	34	3	33	24
	28	700	905	840	800	34	3	33	24
	30	750	970	900	855	36	3	33	24
	32	800	1020	950	905	36	3	33	28
	36	900	1120	1050	1005	38	3	33	28
	40	1000	1235	1160	1110	40	3	39	28
	44	1100	1345	1270	1220	42	3	39	28
	48	1200	1465	1380	1325	44	3	39	32
	54	1350	1630	1540	1480	48	3	45	36
	60	1400	1795	1700	1635	50	3	45	40

Pipe Flange - JIS B2220 RF / RF



Class	NPS		Ø D	Ø D1	Ø D2	b	f	Bolt	
	inch	mm						Ø d	Number
16K	1/2	15	95	70	51	12	1	15	4
	3/4	20	100	75	56	14	1	15	4
	1	25	125	90	67	14	1	19	4
	1-1/4	32	135	100	76	16	2	19	4
	1-1/2	40	140	105	81	16	2	19	4
	2	50	155	120	96	16	2	19	8
	2-1/2	65	175	140	116	18	2	19	8
	3	80	200	160	132	20	2	23	8
	4	100	225	185	160	22	2	23	8
	5	125	270	225	195	22	2	25	8
	6	150	305	260	230	24	2	25	12
	8	200	350	305	275	26	2	25	12
	10	250	430	380	345	28	2	27	12
	12	300	480	430	395	30	3	27	16
	14	350	540	480	440	34	3	33	16
	16	400	605	540	495	38	3	33	16
	18	450	675	605	560	40	3	33	20
	20	500	730	660	615	42	3	33	20
	22	550	795	720	670	44	3	39	20
	24	600	845	770	720	46	3	39	24
	26	650	895	820	770	48	5	39	24
	28	700	960	875	820	50	5	42	24
	30	750	1020	935	880	52	5	42	24
	32	800	1085	990	930	54	5	48	24
	36	900	1185	1090	1030	58	5	48	28
20K	1/2	15	95	70	51	14	1	15	4
	3/4	20	100	75	56	16	1	15	4
	1	25	125	90	67	16	1	19	4
	1-1/4	32	135	100	76	18	2	19	4
	1-1/2	40	140	105	81	18	2	19	4
	2	50	155	120	96	18	2	19	8
	2-1/2	65	175	140	116	20	2	19	8
	3	80	200	160	132	22	2	23	8
	4	100	225	185	160	24	2	23	8
	5	125	270	225	195	26	2	25	8
	6	150	305	260	230	28	2	25	12
	8	200	350	305	275	30	2	25	12
	10	250	430	380	345	34	2	27	12
	12	300	480	430	395	36	3	27	16
	14	350	540	480	440	40	3	33	16
	16	400	605	540	495	46	3	33	16
	18	450	675	605	560	48	3	33	20
	20	500	730	660	615	50	3	33	20
	22	550	795	720	670	52	3	39	20
	24	600	845	770	720	54	3	39	24
	26	650	945	850	790	60	5	48	24
	28	700	995	900	840	64	5	48	24
	30	750	1080	970	900	68	5	56	24
	32	800	1140	1030	960	72	5	56	24
	36	900	1250	1140	1070	76	5	56	28

Pipe Flange - JIS B2220 RF / FF



Class	NPS		Ø D	Ø D1	Ø D2	b	f	Bolt		
	inch	mm						Ø d mm	Number	Dia.
40K	1/2	15	115	80	55	20	1	19	4	M16
	3/4	20	120	85	60	20	1	19	4	M16
	1	25	130	95	70	22	1	19	4	M16
	1-1/4	32	140	105	80	24	2	19	4	M16
	1-1/2	40	160	120	90	24	2	23	4	M20
	2	50	165	130	105	26	2	19	8	M16
	2-1/2	65	200	160	130	30	2	23	8	M20
	3	80	210	170	140	32	2	23	8	M20
	4	100	250	205	165	36	2	25	8	M22
	5	125	300	250	200	40	2	27	8	M24
	6	150	355	295	240	44	2	33	12	M30
	8	200	405	345	290	50	2	33	12	M30
	10	250	475	410	355	56	2	33	12	M30
	12	300	540	470	410	60	3	39	16	M36
	14	350	585	515	455	64	3	39	16	M36
	16	400	646	570	515	70	3	39	16	M36
63K	1/2	15	120	85	55	23	1	19	4	M16
	3/4	20	135	95	60	25	1	23	4	M20
	1	25	140	100	70	27	1	23	4	M20
	1-1/4	32	150	110	80	30	2	23	4	M20
	1-1/2	40	175	130	90	32	2	25	4	M22
	2	50	185	145	105	34	2	23	8	M20
	2-1/2	65	220	175	130	38	2	25	8	M22
	3	80	230	185	140	40	2	25	8	M22
	4	100	270	220	165	44	2	27	8	M24
	5	125	325	265	200	50	2	33	8	M30
	6	150	365	305	240	54	2	33	12	M30
	8	200	425	360	290	60	2	33	12	M30
	10	250	500	430	355	68	2	39	12	M36
	12	300	560	485	410	77	3	39	16	M36
	14	350	615	530	455	81	3	46	16	M42
	16	400	680	590	515	89	3	46	16	M42

Pipe Flange - Tolerance

Standard	Flange Dimension		Tolerance	
	Symbol	Dimension		
ASME B16.5	Ø D	Outside Diameter	NPS ≤ 24	± 1.6 mm
			NPS > 24	± 3.2 mm
	Ø D1	Diameter of Bolt Circle		± 0.8 mm
	Ø D2	Diameter of Raised Face	f=2.0 mm	± 1.0 mm
			f=7.0 mm	± 0.5 mm
ASME B16.47 Series A MSS SP-44	b	Flange Thickness	NPS ≤ 18	+3.0 mm, -0.0 mm
			NPS ≥ 20	+5.0 mm, -0.0 mm
	Ø d	Diameter of Bolt Hole		± 1.5 mm
	Ø D	Outside Diameter	NPS ≤ 24	± 1.6 mm
			24 < NPS	± 3.2 mm
ASME B16.47 Series B API 605	Ø D1	Diameter of Bolt Circle		± 0.8 mm
	Ø D2	Diameter of Raised Face	f=2.0 mm, NPS ≥ 26	± 2.0 mm
			f=7.0 mm, NPS ≥ 26	± 1.0 mm
	b	Flange Thickness	NPS ≤ 18	± 3.0 mm, - 0.0 mm
			NPS ≥ 20	± 5.0 mm, - 0.0 mm
JIS B2220	Ø d	Diameter of Bolt Hole		± 1.5 mm
	Ø D	Outside Diameter	NPS ≤ 24	± 1.6 mm
			24 < NPS	± 3.2 mm
	Ø D1	Diameter of Bolt Circle		± 0.8 mm
	Ø D2	Diameter of Raised Face	f=2.0 mm	± 1.0 mm
			f=7.0 mm	± 0.5 mm
	b	Flange Thickness	NPS ≤ 18	± 3.0 mm, - 0.0 mm
			NPS ≥ 20	± 5.0 mm, - 0.0 mm
	Ø d	Diameter of Bolt Hole		± 1.5 mm
	Ø D	Outside Diameter	NPS < 600A	± 1.5 mm
			NPS ≥ 600A	± 3.0 mm
	Ø D1	Diameter of Bolt Circle	NPS < 950A	± 0.8 mm
			NPS ≥ 950A	± 1.5 mm
	Ø D2	Diameter of Raised Face	NPS < 700A	± 0.8 mm
			NPS ≥ 700A	± 1.5 mm
	b	Flange Thickness	b < 20.0 mm	+ 1.5, -0.0 mm
			20.0 mm ≤ b < 50.0 mm	+ 2.0, -0.0 mm
			50.0 mm ≤ b	+ 3.0, -0.0 mm
	Ø d	Diameter of Bolt Hole		± 0.8 mm

Standard	RTJ Flange		Tolerance	
	Symbol	Dimension		
ASME B16.5	Ø P	Pitch Diameter		± 0.13 mm
	E	Depth		± 0.4, -0.0 mm
	F	Width		± 0.2 mm
	R	Radius at Bottom	R ≤ 2	+0.8, -0.0 mm
			R > 2	± 0.8 mm
	23 Deg.			± 1/2 Deg.